
UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 2: Systems: From Continuous-Time to Discrete-Time

Lecturer Prof. Justin Haldar

Collected by Gewenxin Yu

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Last Time / Today

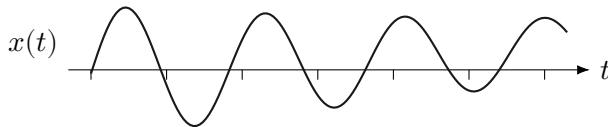
Last time: Defined digital and discrete-time signals.

Today: Systems! Review continuous-time systems, introduce discrete-time systems, and introduce notation and building blocks.

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1 Review: Continuous-Time Systems

Recall a continuous-time signal (1D) is a real- or complex-valued function defined on the real line.



A **continuous-time system** is a mapping from an input signal $x(t)$ to an output $y(t)$:

$$x(t) \rightarrow \boxed{\begin{array}{c} \text{System} \\ H \end{array}} \rightarrow y(t), \quad y(t) = H\{x(t)\}.$$

Example (Continuous-Time Systems).

$$y(t) = x(t),$$

$$y(t) = x(t - \tau),$$

$$y(t) = \int_0^1 x(t - \tau) d\tau,$$

$$y(t) = |x(t)|.$$

3 1.1 Linear Systems

Definition (Homogeneity).

If $H\{x(t)\} = y(t)$, then $H\{\alpha x(t)\} = \alpha y(t)$ for all $\alpha \in \mathbb{C}$ and all $x(t)$.

Definition (Additivity).

If $H\{x_1(t)\} = y_1(t)$ and $H\{x_2(t)\} = y_2(t)$, then $H\{x_1(t) + x_2(t)\} = y_1(t) + y_2(t)$ for all $x_1(t)$ and $x_2(t)$.

1.2 Time-Invariant Systems

Definition (Time-Invariance).

If $H\{x(t)\} = y(t)$, then $H\{x(t - t_0)\} = y(t - t_0)$ for all $t_0 \in \mathbb{R}$ and all $x(t)$.

4 1.3 Linear Time-Invariant Systems (LTI)

LTI systems are characterized by their **impulse response**:

$$\begin{aligned} h(t) &\triangleq H\{\delta(t)\}, \\ H\{\delta(t - t_0)\} &= h(t - t_0). \end{aligned}$$

Notice that $x(t)$ can be decomposed using the sifting property of the delta

function:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau.$$

Applying $H\{\cdot\}$ to both sides and using linearity + time-invariance:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (\textbf{convolution}).$$

5 1.4 Frequency Response

LTI systems are also characterized by their **frequency response** $H(j\Omega)$, the Fourier transform of $h(t)$:

$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt, \quad \Omega \in \mathbb{R}, \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega, \quad t \in \mathbb{R}. \end{aligned}$$

Eigenfunctions of LTI systems. Let $x(t) = e^{j\Omega_0 t}$ with Ω_0 real. Then

$$\begin{aligned} y(t) &= \int h(\tau) x(t - \tau) d\tau \\ &= \int h(\tau) e^{j\Omega_0(t-\tau)} d\tau \\ &= e^{j\Omega_0 t} \int h(\tau) e^{-j\Omega_0 \tau} d\tau \\ &= x(t) H(j\Omega_0). \end{aligned}$$

6 So $x(t) = e^{j\Omega_0 t}$ is an **eigenfunction** of the LTI system, with eigenvalue $H(j\Omega_0)$.
More generally,

$$Y(j\Omega) = H(j\Omega) X(j\Omega).$$

1.5 Transfer Function

LTI systems are also characterized by the **transfer function** $H(s)$, the Laplace transform of $h(t)$:

$$X(s) = \int_{-\infty}^{\infty} e^{-st} x(t) dt, \quad s \in \mathbb{C}, \quad Y(s) = H(s)X(s).$$

- Setting $s = j\Omega$ recovers the Fourier transform.
- Handles more general signals; **regions of convergence (ROC)** tell us about stability and causality.
- Useful for filter design and for solving differential equations.

Example (Solving a Differential Equation with the Laplace Transform).

Consider $\frac{dx}{dt} = -x(t)$. Taking the Laplace transform,

$$sX(s) - x(0) = -X(s) \implies X(s) = \frac{x(0)}{1+s} \implies x(t) = x(0) e^{-t} u(t).$$

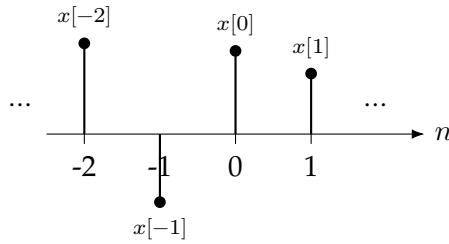
7 2 LTI Discrete-Time Systems (Preview)

Discrete-time LTI systems are characterized analogously:

- Impulse response $h[n]$
- Frequency response $H(e^{j\omega})$: the DTFT
- Transfer function $H(z)$: the z -transform
- Linear (constant-coefficient) difference equations

3 Discrete-Time Signals

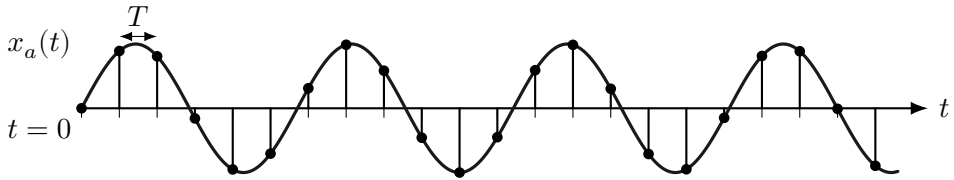
- Represented as a sequence of numbers (“samples”).
- Denoted $x[n]$, where n is an integer, $-\infty < n < \infty$.
- $x[n]$ is **not defined** for non-integer n .



Notation: $\{\dots, x[-2], x[-1], x[0], x[1], \dots\}$, with $n = 0$ at $x[0]$.

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3.1 Sampling of Continuous-Time Signals



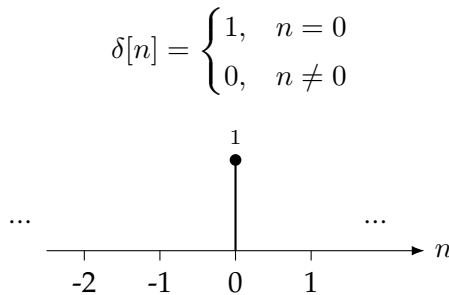
$$x[n] = x_a(t)|_{t=nT} = x_a(nT).$$

T : **sampling interval / sampling period.**

$f_T = \frac{1}{T}$: **sampling rate / sampling frequency.**

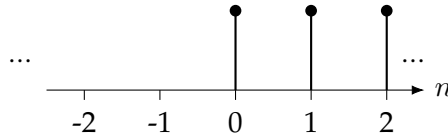
3.2 Basic Sequences

Unit impulse sequence



Unit step sequence

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



9 3.3 Real Sinusoidal Sequences

$$x[n] = A \cos(\omega_0 n + \phi),$$

where A is the amplitude,
 ω_0 is the frequency, and ϕ is the phase.

3.4 Complex (Damped) Sinusoids

$$x[n] = A\alpha^n, \quad -\infty < n < \infty.$$

A and α can be real or complex.

Write $\alpha = e^{\sigma_0} e^{j\omega_0}$ (σ_0, ω_0 real) and $A = |A| e^{j\phi}$. Then

$$x[n] = |A| e^{j\phi} e^{(\sigma_0 + j\omega_0)n} = x_{re}[n] + j x_{im}[n],$$

$$x_{re}[n] = |A| e^{\sigma_0 n} \cos(\omega_0 n + \phi),$$

$$x_{im}[n] = |A| e^{\sigma_0 n} \sin(\omega_0 n + \phi).$$

- $\sigma_0 = 0 \Rightarrow$ constant amplitude
- $\sigma_0 < 0 \Rightarrow$ exponential decay
- $\sigma_0 > 0 \Rightarrow$ exponential growth

10 3.5 Arbitrary Sequences can be represented as a weighted sum of time-shifted versions of basic sequences

Example.

$x[n] = \{0.5, 1, -1, 0.5\}$ with $n = 0$ at the sample 1 can be written as

$$x[n] = 0.5 \delta[n + 1] + \delta[n] - \delta[n - 1] + 0.5 \delta[n - 2],$$

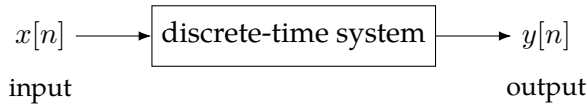
or, in terms of shifted unit steps,

$$x[n] = 0.5 u[n + 1] + 0.5 u[n] - 2 u[n - 1] + 1.5 u[n - 2] - 0.5 u[n - 3].$$

Fact.

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n - m], \quad \text{e.g.} \quad u[n] = \sum_{m=0}^{\infty} \delta[n - m].$$

4 Discrete-Time Systems



11 4.1 Basic Operations

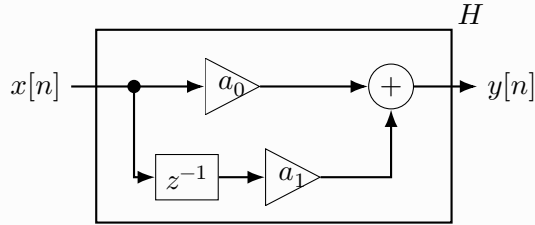
- **Multiplication (scalar):** $x[n] \rightarrow \boxed{A} \rightarrow y[n]$, $y[n] = A x[n]$.
- **Delay:** $x[n] \rightarrow \boxed{z^{-1}} \rightarrow y[n]$, $y[n] = x[n - 1]$.
- **Time reversal:** $x[n] \rightarrow \boxed{\text{time reversal}} \rightarrow y[n]$, $y[n] = x[-n]$.
- **Advance:** $x[n] \rightarrow \boxed{z} \rightarrow y[n]$, $y[n] = x[n + 1]$.
- **Branching:** $x[n] \rightarrow \bullet \rightarrow \begin{matrix} x[n] \\ x[n] \end{matrix}$ $x[n]$ splits into two identical copies.

- **Addition:** $x[n] \rightarrow \bigoplus \rightarrow y[n] = x[n] + w[n]$

- **Product (modulation):** $x[n] \longrightarrow \bigcirc \times \longrightarrow y[n] = x[n]w[n]$
 $\uparrow$
 $w[n]$

Example.

A two-tap system:



$$y[n] = a_0 x[n] + a_1 x[n - 1].$$

Its impulse response is

$$h[n] = H\{\delta[n]\} = a_0 \delta[n] + a_1 \delta[n - 1] = \{a_0, a_1\}, \quad n = 0 \text{ at } a_0.$$

This is an **FIR filter** (Finite Impulse Response).

Example (Moving-Average System).

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k].$$

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 4.1–4.4
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 1.4